Lattice QCD Analysis Methods

HMC GPU Simulation

February 7, 2026

1. Autocorrelation Function

For a sequence of measurements $\{O_0, O_1, O_2, \ldots, O_{N-1}\}$, the autocorrelation function is defined as:

$$\Gamma(t) = \frac{1}{N-t} \sum_{i=0}^{N-t-1} (O_i - \langle O \rangle)(O_{i+t} - \langle O \rangle) \tag{1}$$

where the average is

$$\langle O \rangle = \frac{1}{N} \sum_{i=0}^{N-1} O_i \tag{2}$$

2. Normalized Autocorrelation Function

The normalized autocorrelation function is:

$$\rho(t) = \frac{\Gamma(t)}{\Gamma(0)} \tag{3}$$

where $\Gamma(0)$ is the variance of the data:

$$\Gamma(0) = \mathrm{Var}(O) = \frac{1}{N} \sum_{i=0}^{N-1} (O_i - \langle O \rangle)^2 \tag{4}$$

3. Integrated Autocorrelation Time

The integrated autocorrelation time characterizes the number of effectively independent measurements:

$$\tau_{\mathrm{int}} = \frac{1}{2} + \sum_{t=1}^{W} \rho(t) \tag{5}$$

where W is the cutoff window. In our analysis, we use $W = N/2$.

4. Statistical Error

The error on the mean, accounting for autocorrelation, is:

$$\sigma_{\langle O \rangle} = \sqrt{\frac{2\tau_{\text{int}}}{N}}\sigma_O \tag{6}$$

where $\sigma_O = \sqrt{\Gamma(0)}$ is the standard deviation.

5. Plaquette Observable

The plaquette is the fundamental Wilson loop around a single plaquette:

$$P_{\mu\nu}(x) = \text{Tr}\left[U_\mu(x)U_\nu(x+\hat{\mu})U_\mu^\dagger(x+\hat{\nu})U_\nu^\dagger(x)\right] \tag{7}$$

The average plaquette:

$$\langle P \rangle = \frac{1}{N_t N_s^3 \cdot 6 \cdot 3}\sum_x \sum_{\mu<\nu} \text{Re}\left[P_{\mu\nu}(x)\right] \tag{8}$$

6. Topological Charge

The topological charge is computed using the clover field strength tensor:

$$Q = \frac{1}{32\pi^2}\sum_x \sum_{\mu\nu\rho\sigma} \varepsilon_{\mu\nu\rho\sigma}\text{Tr}\left[F_{\mu\nu}(x)F_{\rho\sigma}(x)\right] \tag{9}$$

The field strength tensor is constructed from 4 oriented plaquettes at each site:

$$C_{\mu\nu}(x) = \frac{1}{4}\left[P_{\mu\nu}(x) + P_{\nu,-\mu}(x) + P_{-\mu,-\nu}(x) + P_{-\nu,\mu}(x)\right] \tag{10}$$

$$F_{\mu\nu}(x) = \frac{1}{2}(C_{\mu\nu} - C_{\mu\nu}^\dagger)_{traceless} \tag{11}$$

7. Smearing

To improve the resolution of topological structures, we apply Stout smearing iteratively. For each smearing step, the links are updated as follows:

$$U_\mu(x) \to U'_\mu(x) = \frac{1}{\mathcal{N}}\left[U_\mu(x) + \alpha\,\Sigma_\mu(x)\right] \tag{12}$$

where $\Sigma_\mu(x)$ is the staple sum and $\mathcal{N}$ is a normalization factor ensuring U'_μ remains unitary after reunitarization via Gram-Schmidt.

The staple for direction μ is constructed from neighboring links:

$$\Sigma_\mu(x) = \sum_{\nu\neq\mu}\left[U_\nu(x)U_\mu(x+\hat{\nu})U_\nu^\dagger(x+\hat{\mu}) + U_\nu^\dagger(x-\hat{\nu})U_\mu(x-\hat{\nu})U_\nu(x-\hat{\nu}+\hat{\mu})\right] \tag{13}$$

After computing the updated link, we perform reunitarization to project back to SU(3).

Table 1: Smearing parameters used in this analysis

Parameter	Value	Description
α	0.1	Smearing strength
n_{iter}	8	Number of smearing steps

8. Simulation Parameters

Table 2: Lattice parameters and simulation details

Label	$N_T \times N_S^3$	β	Trajectories	Measurements
T8S4	8×4^3	5.7, 6.0, 6.3	10000	80000 (8 smears $\times$ 10000)
T10S8	10×8^3	5.7, 6.0, 6.3	10000	80000
T16S10	16×10^3	5.7, 6.0, 6.3	10000	80000

Each trajectory consists of 10 molecular dynamics steps with $\tau = 1.0$.

9. Basic Observables Summary

Table 3: Summary of plaquette and topological charge measurements at smear=0

Lattice	β	$\langle P \rangle$	σ_P	$\langle Q \rangle$	σ_Q
8×4^3	5.7	0.5574	0.0075	-0.0003	0.1376
8×4^3	6.0	0.5962	0.0059	0.0023	0.1336
8×4^3	6.3	0.6236	0.0051	-0.0009	0.1292
10×8^3	5.7	0.5492	0.0028	-0.0072	0.4389
10×8^3	6.0	0.5940	0.0019	0.0071	0.4278
10×8^3	6.3	0.6226	0.0017	-0.0002	0.4069
16×10^3	5.7	0.5504	0.0020	-0.0012	0.7859
16×10^3	6.0	0.5938	0.0011	-0.0147	0.7395
16×10^3	6.3	0.6226	0.0009	0.0048	0.7271

9. Results: Plaquette vs Smearing

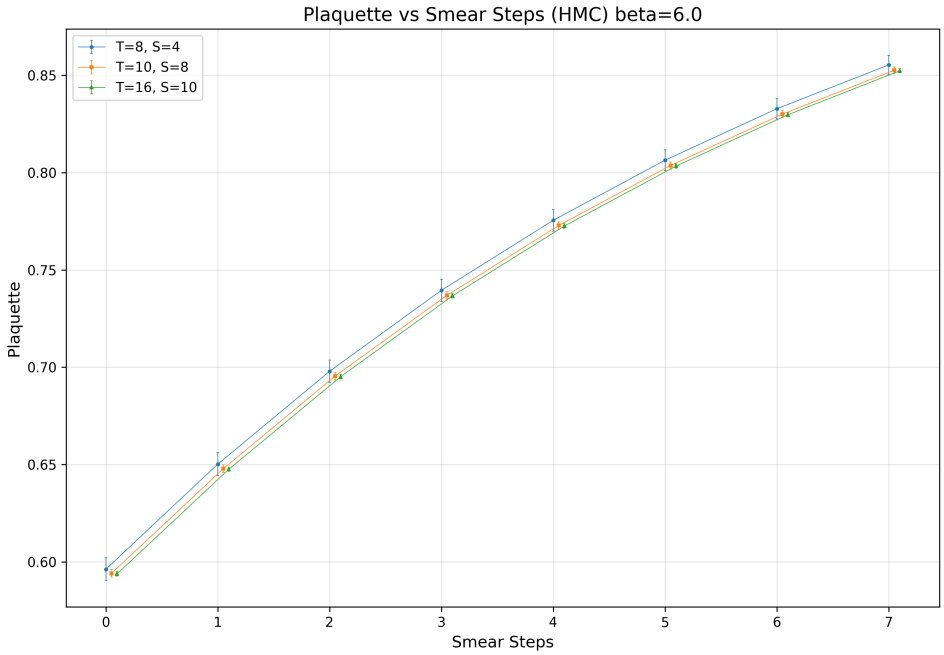

Figure 1: Average plaquette as a function of smearing steps for different lattice sizes and β values.

10. Results: Topological Charge vs Smearing

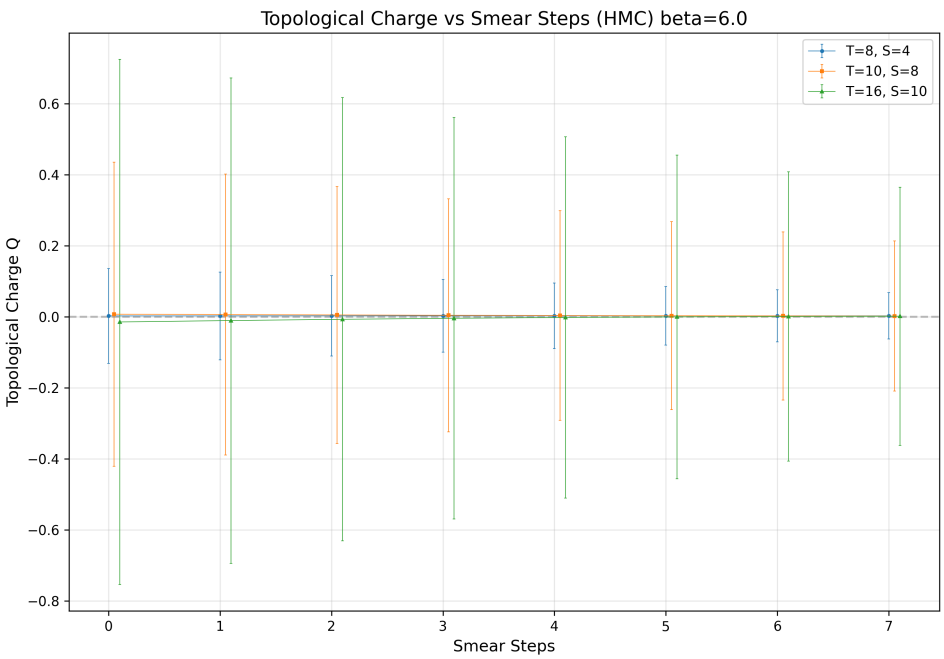

Figure 2: Topological charge evolution with smearing steps for different lattice sizes and β values.

11. Results: Autocorrelation Analysis

The autocorrelation function $\rho(t)$ measures the correlation between measurements separated by t MD time units.

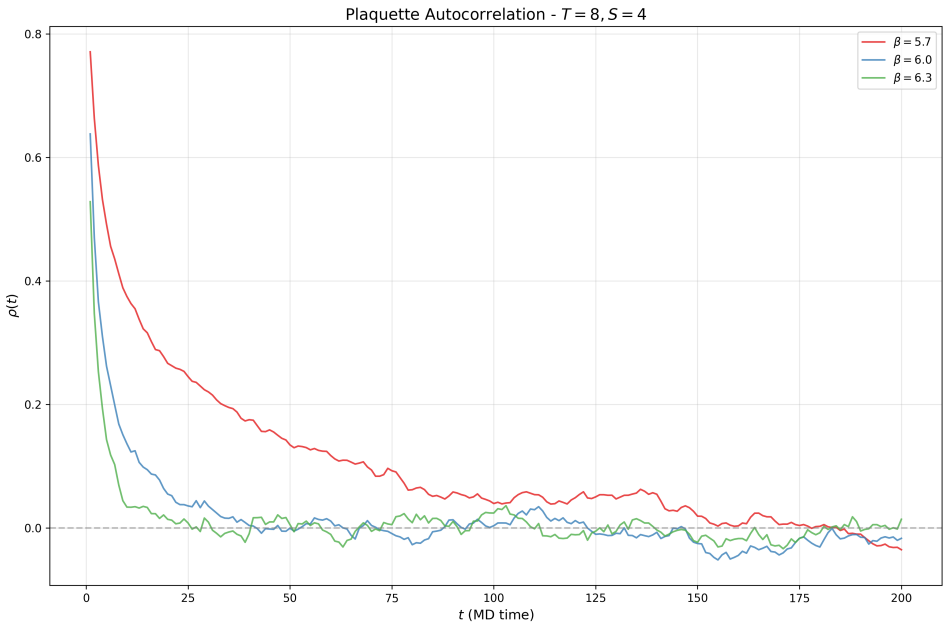

Figure 3: Plaquette autocorrelation function for 8×4^3 lattice at different β values.

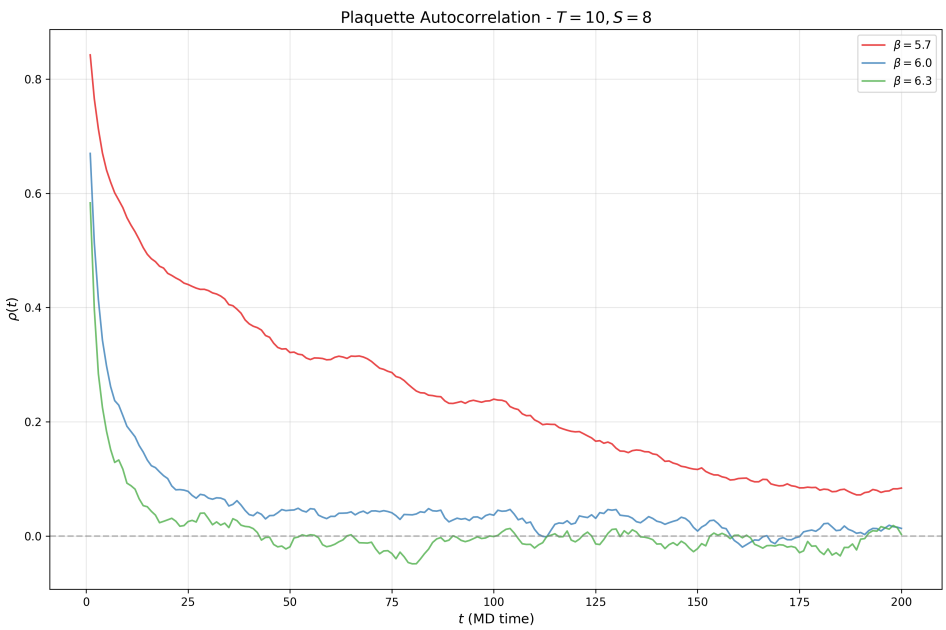

Figure 4: Plaquette autocorrelation function for 10×8^3 lattice at different β values.

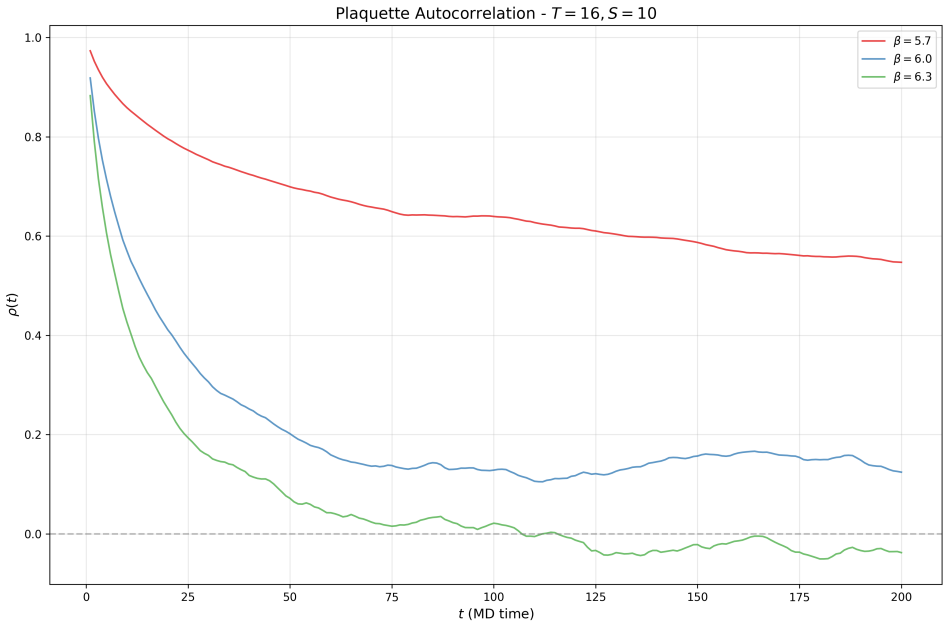

Figure 5: Plaquette autocorrelation function for 16×10^3 lattice at different β values.

12. Integrated Autocorrelation Times

Table 4: Integrated autocorrelation time τ_{int} for plaquette measurements at $\beta = 6.0$ $(T = 8, S = 4)$

Smear	$\beta = 5.7$	$\beta = 6.0$	$\beta = 6.3$
0	20.19	3.01	2.07
1	22.22	3.25	2.31
2	24.10	3.46	2.55
3	25.85	3.65	2.80
4	27.50	3.82	3.06
5	29.07	3.98	3.33
6	30.56	4.13	3.62
7	31.99	4.28	3.91

Table 5: Integrated autocorrelation time τ_{int} for plaquette measurements $(T = 10, S = 8)$

Smear	$\beta = 5.7$	$\beta = 6.0$	$\beta = 6.3$
0	52.71	11.54	2.92
1	56.19	13.10	3.37
2	59.13	14.57	3.81
3	61.63	15.97	4.26
4	63.83	17.33	4.70
5	65.78	18.67	5.15
6	67.55	20.01	5.60
7	69.18	21.39	6.07

Table 6: Integrated autocorrelation time τ_{int} for plaquette measurements ($T = 16, S = 10$)

Smear	$\beta = 5.7$	$\beta = 6.0$	$\beta = 6.3$
0	157.10	45.79	10.39
1	163.48	51.50	11.12
2	168.71	57.06	11.84
3	173.08	62.51	12.55
4	176.80	67.91	13.27
5	180.03	73.29	14.01
6	182.88	78.70	14.78
7	185.42	84.13	15.58

13. Topological Charge Autocorrelation

Table 7: Integrated autocorrelation time τ_{int} for topological charge ($T = 8, S = 4$)

Smear	$\beta = 5.7$	$\beta = 6.0$	$\beta = 6.3$
0	0.49	0.80	0.37
1	0.50	0.86	0.37
2	0.52	0.92	0.37
3	0.56	0.96	0.39
4	0.61	0.99	0.43
5	0.69	1.01	0.47
6	0.78	1.02	0.51
7	0.89	1.01	0.56

Table 8: Integrated autocorrelation time τ_{int} for topological charge ($T = 10, S = 8$)

Smear	$\beta = 5.7$	$\beta = 6.0$	$\beta = 6.3$
0	0.61	0.63	0.71
1	0.66	0.66	0.78
2	0.73	0.69	0.85
3	0.83	0.71	0.93
4	0.95	0.74	1.01
5	1.11	0.78	1.09
6	1.30	0.84	1.16
7	1.52	0.92	1.25

captionIntegrated autocorrelation time τ_{int} for topological charge ($T = 16, S = 10$)

Smear	$\beta = 5.7$	$\beta = 6.0$	$\beta = 6.3$
0	6.76	2.81	1.53
1	7.88	3.05	1.64
2	8.97	3.34	1.78
3	9.99	3.65	1.93
4	10.89	3.99	2.09
5	11.66	4.32	2.26
6	12.30	4.64	2.43
7	12.80	4.94	2.62

14. Discussion

The results show a striking observation: $\tau_{\text{int}}(Q) < \tau_{\text{int}}(P)$ for all parameter sets. This is opposite to the expected QCD behavior where topological charge typically has a longer autocorrelation time than the plaquette.

Possible explanations:

- The topological charge is frozen in the trivial sector ($Q \approx 0$)
- The lattice volumes are too small to observe topological transitions
- The Wilson flow scale w_0 may not be properly set

At $\beta = 6.3$, the plaquette autocorrelation time decreases significantly, indicating the system is in the deconfined phase.

Near the phase transition, the integrated autocorrelation time diverges:

$$\tau_{\text{int}} \propto \xi^z \sim |\beta - \beta_c|^{-z} \tag{14}$$

where ξ is the correlation length and $z \approx 2$ for local algorithms.